

Synthetic Market Phase 13

Portfolio-ladder geometry on the same 4-asset market frame used in Phases 9-12. This phase asks whether a second context family deforms the shared market coordinates in the same way as risk, or whether portfolio context behaves more locally and asset-relatively.

DATE	BASE DATA	STATES	OBJECT
22 March 2026	Phase 13 portfolio ladder	row_mean L4, row_eos L14	adjacent portfolio-step transforms

MAIN READ

Phase 13 shows that portfolio context is not just another copy of the risk story. The shared 4-asset market frame still survives almost intact early, but later portfolio effects are best described as local reallocations around that frame rather than as one strong end-to-end global warp.

Early row_mean @ L4 coordinate transfer stays above 0.989 R^2 for both latent axes across portfolio_1..portfolio_5. Later row_eos @ L14 states still realign toward portfolio-adjusted score geometry, but the transform structure is middle-heavy: portfolio_1 -> portfolio_2, portfolio_2 -> portfolio_3, and portfolio_3 -> portfolio_4 admit strong local fits, while market_only -> portfolio_1 and market_only -> portfolio_5 remain much weaker.

EARLY FRAME FLOOR	LATE STRONGEST LOCAL FIT	LATE END-TO-END	NONTRIVIAL COMPOSE
0.98921 latent_x, market->portfolio_4	0.94023 diagonal on P2-P3	0.49596 identity on M-P5	0.99961 orthogonal matrix cosine

Why Phase 13 Exists

Phase 12 established a strong risk result: the same 4-asset market geometry survives the DX-native risk ladder, and the late end-to-end ladder still composes most coherently under near-rigid maps. The next question was whether that was just a “settings in general” story or something specific to risk.

Portfolio is a useful second family because it is qualitatively different. In the synthetic prompts:

- free ETH decreases across the ladder
- one asset is already held
- the prompt increasingly warns against concentration

That should create an asset-relative overlay. Unlike risk, this context is not supposed to apply the same penalty to every asset.

Experimental Design

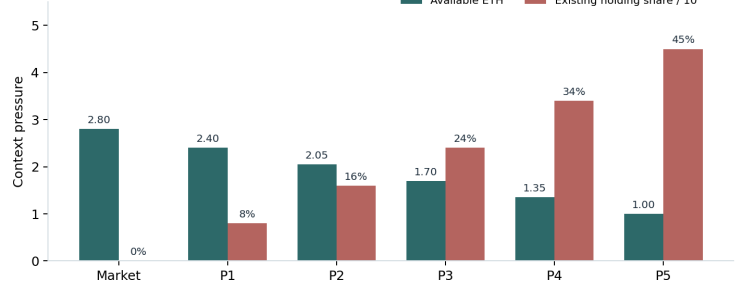
Phase 13 dataset design and experimental target

What data was used

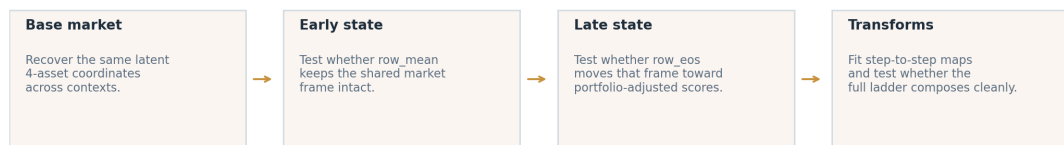
The dataset is built from repeated versions of the same 4-asset markets, not from one-off prompts.

1. 4 latent market scenarios
 2. 2 surface styles
 3. 4 row/symbol permutations
 4. 3 global magnitude scales
 5. 6 portfolio contexts
- Total prompts: 576
Asset rows: 2304
Pairwise rows: 6912

What a portfolio ladder means in this dataset



What is being tested



Phase 13 is built from repeated variants of the same 4-asset markets. The figure shows the dataset recipe, what the portfolio ladder means, and the actual question being tested.

The easiest way to read this phase is:

- start with a base 4-asset market
- render many nuisance variants of the same market
- place each variant into six contexts: `market_only`, then `portfolio_1..portfolio_5`
- ask whether the model keeps the same market coordinates and how those coordinates change across the ladder

Concretely, the dataset contains:

- 4 latent market scenarios
- 2 surface styles
- 4 row / symbol permutations
- 3 global scale variants
- 6 contexts per base market

That yields:

- 576 prompts total
- 2,304 asset rows
- 6,912 pairwise asset comparisons

In this report, the surface styles, row permutations, and global magnitude scales are all *nuisance variants*. That means they change how the same market is written on the page without changing the underlying latent market shape we actually care about.

What “Portfolio Ladder” Means

In this report, a “ladder” just means an ordered sequence of matched prompt variants. The market stays the same, but the contextual pressure changes step by step.

For Phase 13:

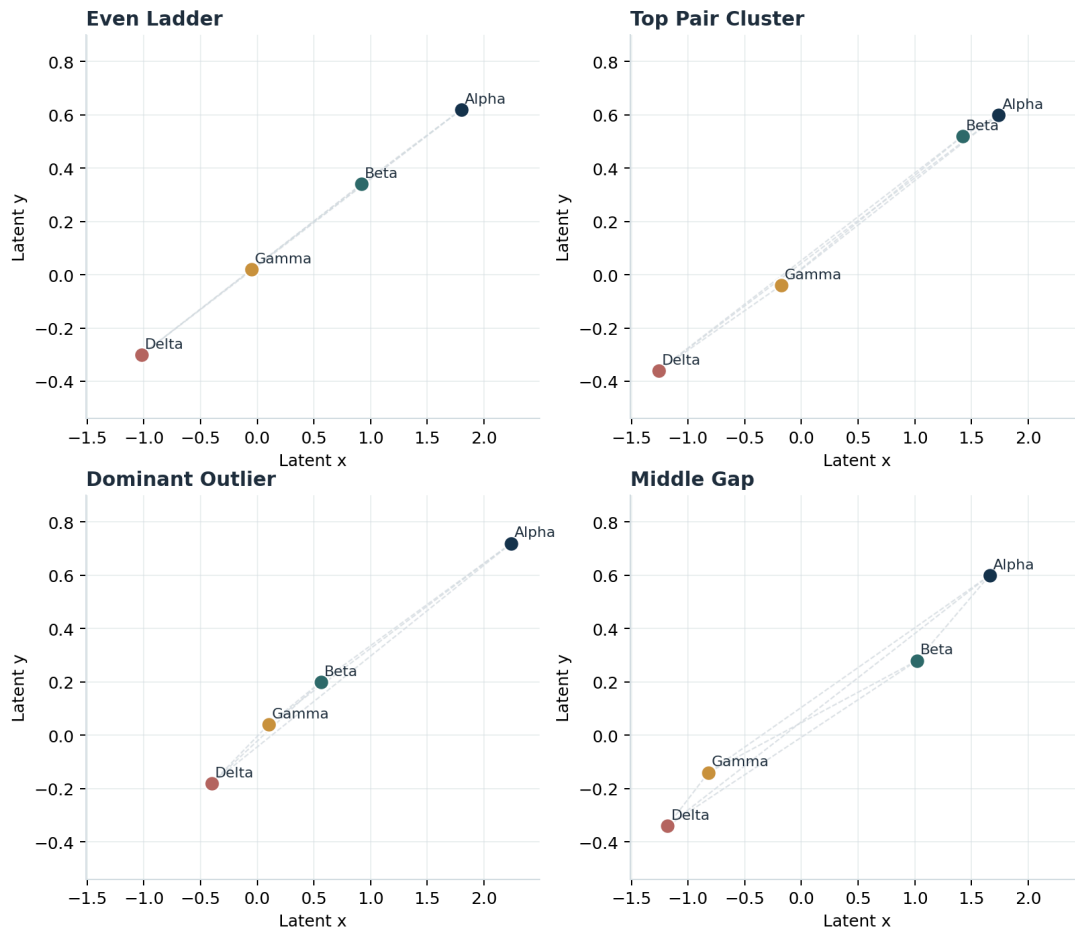
- market_only has no portfolio overlay
- portfolio_1 introduces a small existing position and plenty of free ETH
- portfolio_5 has the largest existing position, the least free ETH, and the strongest anti-concentration instruction

So the ladder is not a time series. It is a controlled context sweep over the same base market.

Context	Available ETH	Existing position	Instructional effect
market_only	2.80	none	No portfolio-specific pressure.
portfolio_1	2.40	8%	Light pressure against adding more concentration.
portfolio_2	2.05	16%	Moderate diversification pressure begins.
portfolio_3	1.70	24%	The held asset becomes more expensive to add to.
portfolio_4	1.35	34%	Concentration avoidance becomes strong.
portfolio_5	1.00	45%	Strongest reallocation pressure toward alternatives.

What “4-Asset Market Geometry” Means

The 4-asset geometry object: same assets, different whole-market shapes



Each panel is a different latent market shape. The goal is not just to recover a winner, but to preserve the relative placement of all four assets at once.

The four latent market-shape families used in the dataset. Each point is one asset. Dashed edges mark the pairwise distances that collectively define the whole-market geometry.

The key object here is not a single asset row and not a single pairwise comparison. It is the *shape of the 4-asset market*.

That means:

- four assets are placed at known latent coordinates
- those coordinates define six pairwise distances
- the distances together determine the market’s overall shape

Why this matters:

- two markets can have the same winner but different shapes
- two markets can have the same rank order but different spacing or clustering
- a real set-level representation should preserve more than “which token looks best”

So when the report says “shared market frame,” it means a reusable coordinate system that organizes all four assets together.

Scenario name	Plain-language meaning
even ladder	The four assets step down in fairly regular intervals. There is a clear ordering, but no special clustering.
top pair cluster	The top two assets sit close together while the lower two are farther away.
dominant outlier	One asset is clearly separated above the other three, which are comparatively compressed.
middle gap	There is a larger-than-usual split between the upper pair and lower pair of assets.

Plain-Language Terminology

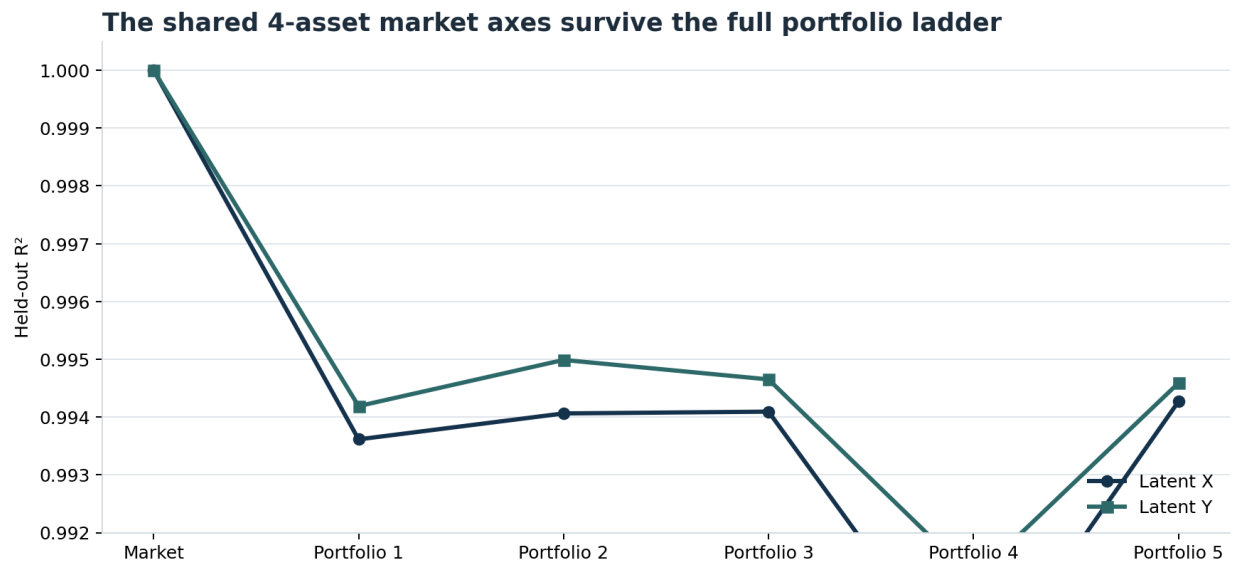
This is the shortest way to decode the jargon used later in the report:

- market frame the shared coordinate system the model seems to use to organize all four assets
- market coordinates the x/y position of each individual asset inside that shared frame
- market geometry the whole shape formed by those four coordinates together
- rigid map a transform that mostly keeps distances and angles intact, like a rotation or reflection
- context-adjusted score geometry the asset layout implied after we rescore the same market under a specific context, such as portfolio pressure
- nuisance variants prompt differences that should not change the underlying market meaning, like row order, symbol aliases, or formatting style

The point of Phase 13 is to separate:

- real geometry changes caused by context
- from fake changes caused by nuisance presentation differences

The Shared Market Frame Still Survives the Full Ladder



Held-out coordinate transfer from `market_only` into each portfolio context. Both latent axes stay near ceiling, with the best readout always in early `row_mean`.

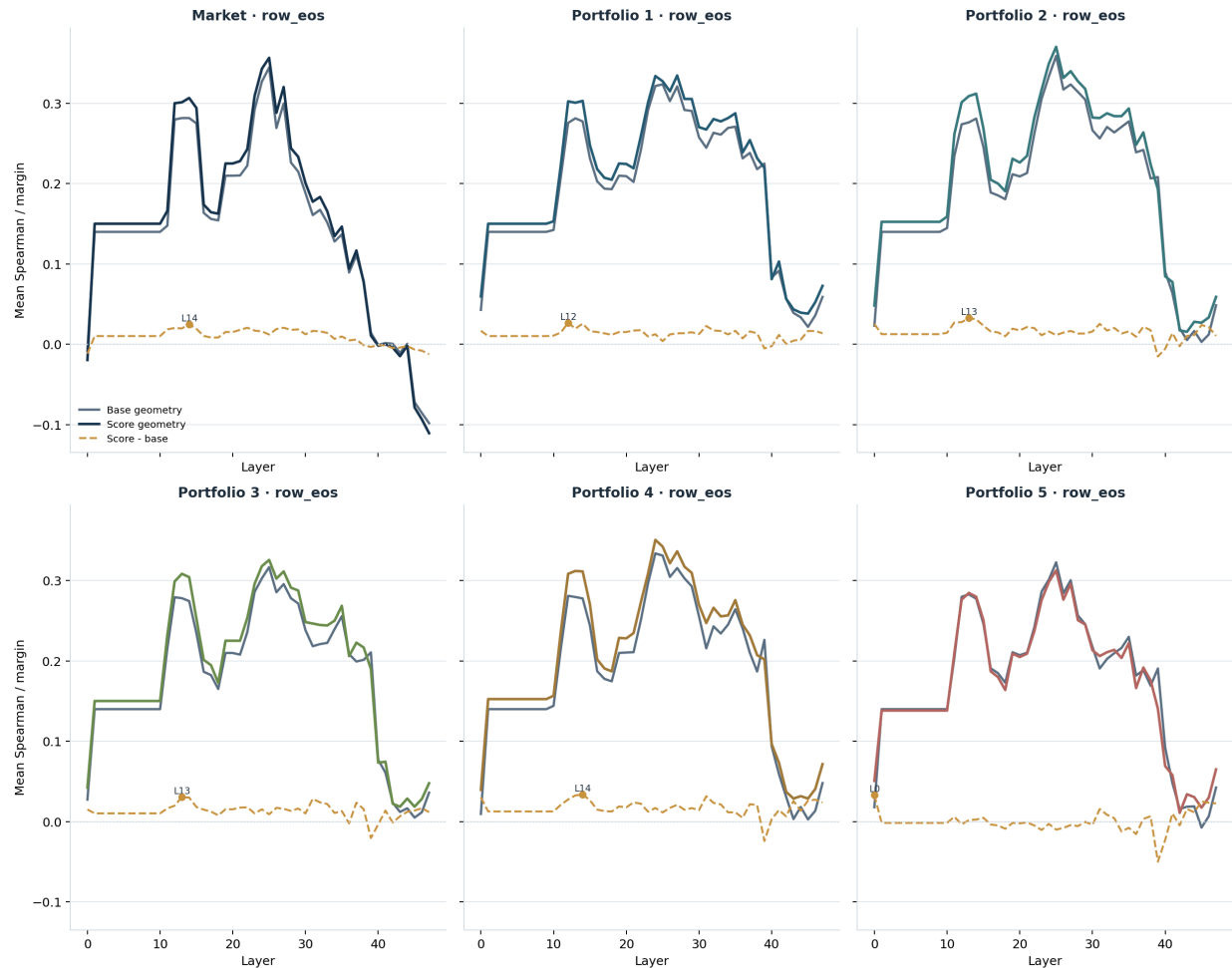
This is the conservative base result, and it is strong:

- `market_only` -> `portfolio_1`: 0.9936 / 0.9942 for `latent_x` / `latent_y`
- `market_only` -> `portfolio_2`: 0.9941 / 0.9950
- `market_only` -> `portfolio_3`: 0.9941 / 0.9947
- `market_only` -> `portfolio_4`: 0.9892 / 0.9912
- `market_only` -> `portfolio_5`: 0.9943 / 0.9946

So portfolio context does *not* erase the model's underlying market coordinates. The same 4-asset frame is still present and linearly recoverable very early.

Late States Still Realign Toward Portfolio-Adjusted Score Geometry

Later row_eos states shift toward context-adjusted score geometry across the portfolio ladder



Realignment margin between recovered activation-space geometry and portfolio-adjusted score geometry for each context.

As in the risk ladder, the later state moves toward the context-adjusted score geometry:

- portfolio_1: margin 0.0268 at row_eos @ L12
- portfolio_2: margin 0.0327 at row_eos @ L13
- portfolio_3: margin 0.0304 at row_eos @ L13
- portfolio_4: margin 0.0337 at row_eos @ L14

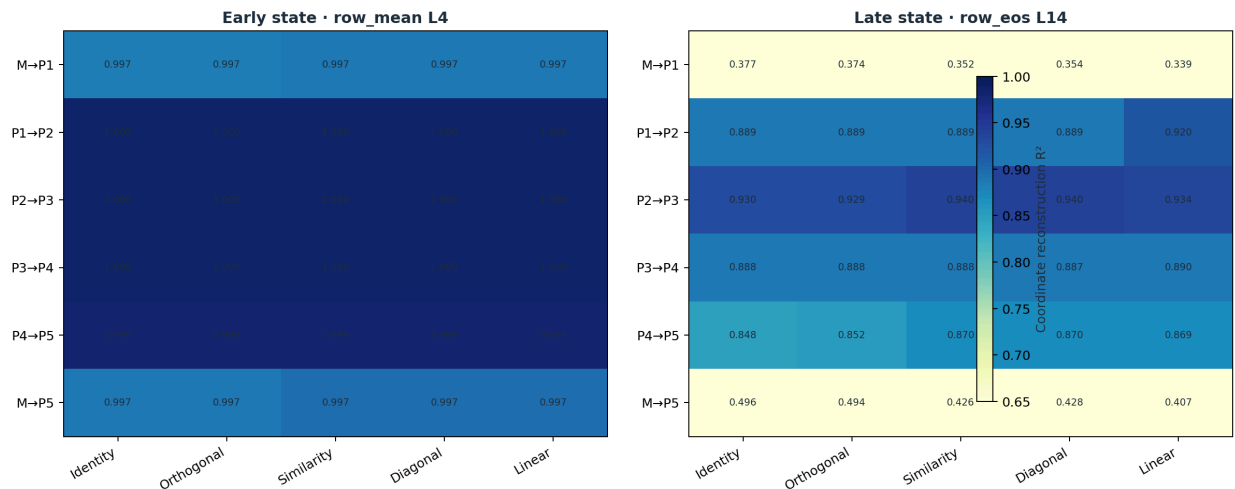
That is a stable late-depth pattern. The one exception is portfolio_5, which numerically peaks at layer 0 but with low NN accuracy (0.0506). That looks unstable and should not be treated as meaningful late structure.

The useful reading is therefore:

- early row_mean preserves the shared market frame
- later row_eos moves that frame toward portfolio-adjusted scores

Early Portfolio Transforms Are Nearly Trivial

Portfolio-step transforms reveal which local maps stay near-rigid and which do not



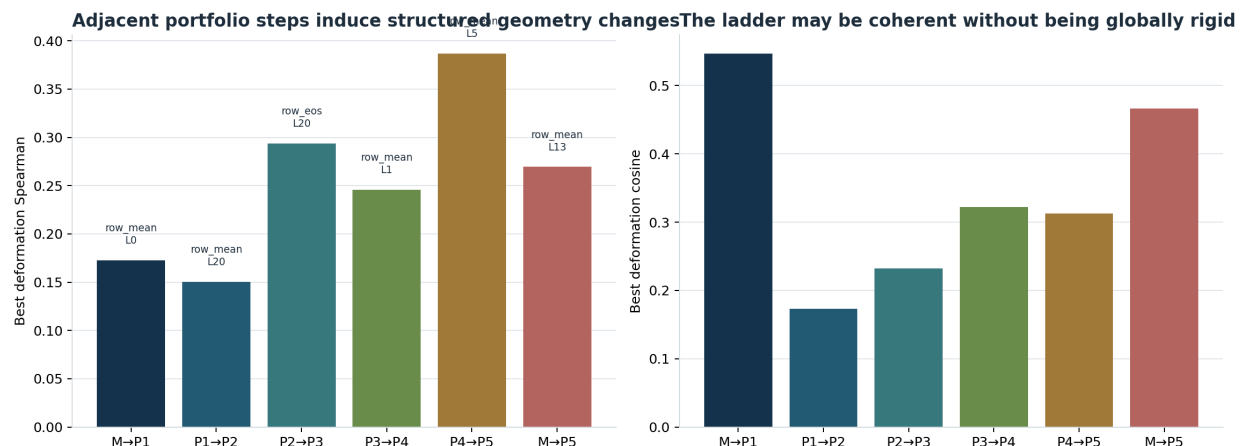
Coordinate reconstruction R^2 for transform families fit on each adjacent portfolio step. Left: early row_mean @ L4. Right: late row_eos @ L14.

The early panel is almost flat in the same way that the early risk panel was. All adjacent-step fits are already near ceiling:

- market_only -> portfolio_1: similarity best, $R^2 = 0.9969$
- portfolio_1 -> portfolio_2: linear best, $R^2 = 0.9996$
- portfolio_2 -> portfolio_3: linear best, $R^2 = 0.9996$
- portfolio_3 -> portfolio_4: similarity best, $R^2 = 0.9996$
- portfolio_4 -> portfolio_5: similarity best, $R^2 = 0.9994$
- market_only -> portfolio_5: similarity best, $R^2 = 0.9970$

This is the same structural point as before: context insertion is not destroying the market coordinates in the early state.

The Late Ladder Is Local, Mixed, and Middle-Heavy



Best deformation margins for adjacent and end-to-end portfolio comparisons. Higher margins indicate stronger evidence that the geometry is shifting toward the context-adjusted arrangement.

The late transform structure is where portfolio diverges from risk. At row_eos @ L14, the strongest local fits sit in the middle of the ladder:

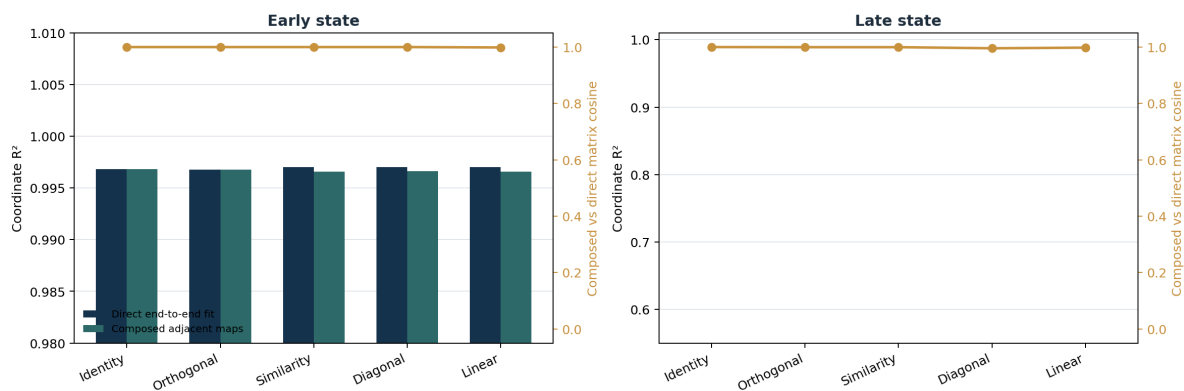
- market_only -> portfolio_1: identity best, $R^2 = 0.377$
- portfolio_1 -> portfolio_2: linear best, $R^2 = 0.920$
- portfolio_2 -> portfolio_3: diagonal best, $R^2 = 0.940$
- portfolio_3 -> portfolio_4: linear best, $R^2 = 0.890$
- portfolio_4 -> portfolio_5: similarity best, $R^2 = 0.870$
- market_only -> portfolio_5: identity best, $R^2 = 0.496$

That pattern matters. The ladder has clean local structure once the portfolio overlay is active, but entering the ladder and the end-to-end market_only -> portfolio_5 jump do not admit one comparably strong global map.

This is exactly what we would expect if portfolio context mostly changes *relative allocation pressure* inside a preserved market frame. The overlay is not “global caution.” It is: “you already hold this one, you have less free ETH, and concentration is now more expensive.”

Composition Keeps Orientation Coherent, Not a Strong Global Warp

End-to-end portfolio behavior can be compared directly against the composition of local steps



For market_only -> portfolio_5, compare direct end-to-end fits against the composition of all adjacent portfolio-step maps. Gold marks matrix cosine between the composed and direct transforms.

The composition plot needs one careful note: identity always composes perfectly by construction, so it is not the informative family here. The more useful comparisons are the nontrivial ones:

- orthogonal
 - direct $R^2 = 0.494$
 - composed $R^2 = 0.489$
 - matrix cosine 0.99961
- linear
 - direct $R^2 = 0.407$
 - composed $R^2 = 0.293$
 - matrix cosine 0.99797

So the end-to-end orientation is still highly coherent, but the portfolio ladder does *not* behave like one strong late global warp. Relative to the risk ladder, this is a more local and more heterogeneous deformation story.

Interpretation

SUPPORTED NOW

The model still carries a shared multi-asset market frame through the full portfolio ladder. Later states then apply context-sensitive local reallocations inside that frame, especially in the middle portfolio steps.

NOT SUPPORTED

A single strong end-to-end portfolio warp is not supported. The `portfolio_5` layer-0 realignment anomaly is also not strong enough to treat as a genuine late-stage effect.

The cleanest synthesis after Phase 13 is:

- risk and portfolio both preserve a shared market frame early
- risk looks globally more coherent as a ladder
- portfolio acts more like an asset-relative redistribution overlay

That is a genuinely interesting result. It says the model is not applying one generic “settings transform” to the market. Different context families appear to deform the same underlying coordinates in different ways.

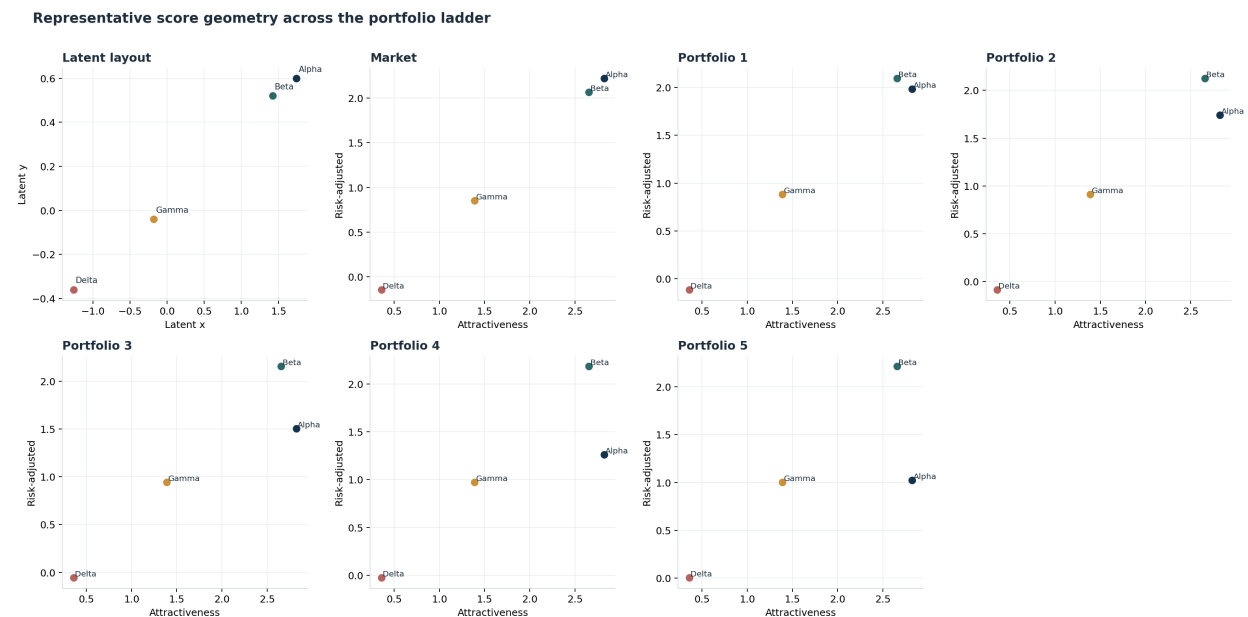
What To Do Next

The next step should keep the same 4-asset geometry object and test one more context family with clearly different semantics:

1. affordance / actionability overlays
2. strategy override overlays

Then bring the same coordinate-and-deformation lens back to a small real-DX matched-variant set. The goal is no longer “does a latent space exist?” The goal is to test whether real contexts also look like structured deformations of a shared market frame.

Appendix A: Representative Portfolio Geometry



Representative `top_pair_cluster` market from the portfolio ladder. The figure shows the same latent base market with portfolio-adjusted score geometries layered on top of it.

This figure anchors what “portfolio deformation” means in the report. The transform analyses above are not abstract regression exercises. They are attempts to recover how the same underlying 4-asset market is reweighted once an existing position, shrinking free ETH, and anti-concentration pressure are added.

Appendix B: Glossary

Term	Meaning
shared market frame	The recovered 2D coordinate system that organizes all four assets before context-specific reweighting.
market coordinates	The individual x/y positions of the four assets inside the shared market frame.
market geometry	The whole-market arrangement of those coordinates, including spacing, clustering, and pairwise distances.
4-asset geometry	The full relative placement of all four assets at once. In practice this means the coordinate layout or, equivalently, the pattern of six pairwise distances between assets.
portfolio ladder	A matched sequence of context variants where the base market is fixed and the portfolio pressure is stepped from <code>market_only</code> to <code>portfolio_5</code> .
nuisance variants	Prompt changes that alter presentation but should not alter the underlying market meaning, such as row order, formatting style, symbol aliases, or global scale formatting.

Term	Meaning
coordinate transfer	How well a probe trained in one context can recover the same latent coordinate in another context. High R^2 means the base frame survived.
realignment margin	How much closer the recovered geometry is to the context-adjusted score geometry than to the raw latent layout. Positive margins mean the context is actively reweighting the market.
context-adjusted score geometry	The geometry implied by rescoreing the same four assets under the current context. In Phase 13, that means incorporating the existing position, shrinking free ETH, and diversification pressure.
identity	No transform at all. If identity already works, the two contexts share almost the same geometry in the decoded frame.
orthogonal	A pure rotation or reflection. Distances and angles are preserved; only orientation changes.
rigid map	A transform that preserves shape up to orientation. In practice here, the closest approximation is an orthogonal map or a near-rigid similarity map.
similarity	Uniform scaling plus rotation. This captures isotropic compression or expansion.
diagonal	Axis-aligned rescaling in the shared coordinate frame. This captures simple non-uniform compression without rotation.
linear	Any 2×2 linear map. This is the most flexible family here and can absorb rotation, anisotropic scaling, and shear-like behavior.
matrix cosine	Cosine similarity between two flattened transform matrices. Here it is used to compare direct end-to-end maps against the composition of adjacent ladder steps.